



Robustness of machine learning to color, size change, normalization, and image enhancement on micrograph datasets with large sample differences

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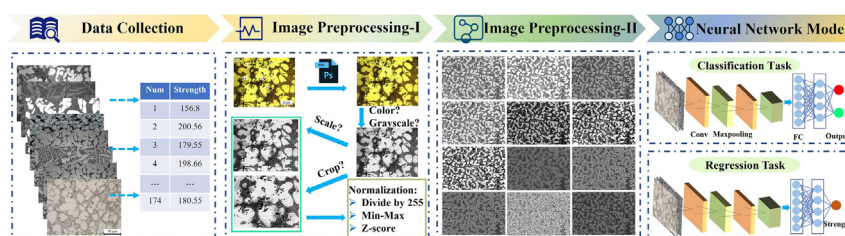
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HIGHLIGHTS

- The impact of various image preprocessing on machine learning using micrograph dataset is evaluated for the first time.
- Color, size variation, normalization and image enhancement are all considered for both classification and regression tasks.
- The impact of preprocessing methods on the machine learn is contingent upon the dataset characteristics and task type.

GRAPHICAL ABSTRACT



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ABSTRACT

Appropriate image preprocessing could improve machine learning performance, but the robustness of machine learning to preprocessing methods in micrograph datasets with significant sample differences remains unexplored. Here we collected hundreds of optical micrographs varied in color, contrast, size and brightness and tensile strength from published literature. After preprocessing including color transformation, size adjustment, normalization and image enhancement, classification model of whether it was texture or not and tensile strength prediction model were established using transfer learning and VGG16 model. Our results showed comparable accuracy between classification models using grayscale and color micrographs. Based on grayscale images, classification model utilizing center cropping achieved an average accuracy 1.09% higher than model employing scaling, while coefficient of determination (R^2) of regression model showed an average increase of 0.0404. The combination of cropping and dividing by 255 was preferable for classification model, yielding an accuracy of 90.79%. Similarly, the combination of cropping and min-max normalization had a better effect on regression model, yielding R^2 of 0.2167. Histogram equalization emerged as the optimal technique for improving classification model, yielding 2.63% increase in accuracy compared to models without image enhancement. For regression models, gamma correction exhibited the highest improvement effect, with an R^2 of 0.2681.

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1. Introduction

Micrographs are crucial data in the field of materials science as they enable researchers to investigate the microstructure of materials and its effect on their properties. Machine learning (ML) techniques have gained popularity in the analysis of micrographs, such as image classification [1], identification and segmentation of material phases [2,3], microstructure reconstruction [4], and material property prediction [5].

High-quality datasets are the foundation for achieving high accuracy rates with ML models. Many of the datasets used in published reports on machine learning are either meticulously prepared or collected under nearly identical processing conditions [1,6–9]. On the other hand, micrographs from other publicly available literature are abundant and highly valuable. Unfortunately, these micrographs vary greatly in color, brightness, contrast, and intensity distribution due to differences in sample preparation, imaging equipment, shooting techniques, and data storage formats.

It has been reported that variations in the micrographs pixel intensity signal caused by instruments can have a negative impact on machine learning models [10]. Tsutsui et al. [11] reported that different imaging equipment can affect the texture characteristics of images, and that a classifier trained using SEM images from one source did not accurately classify SEM images from another source. Kitahara et al. [12] reported that the classification of micrographs on light or dark was a failure. These studies show that using image data of different quality for machine learning often does not achieve the desired results. Appropriate preprocessing of images is a reliable way to improve the performance of machine learning models [13,14].

Image preprocessing typically includes image size changes [15], image normalization [13], image enhancement [14], and more. Image binarization is a straightforward image enhancement technique that is commonly used for simple microstructures [16]. However, when applied to complex micrographs, this method may lead to a significant loss of detail information. Histogram equalization (HE) has been widely utilized to adjust the pixel distribution of images and improve their quality [17]. Bulgarevich et al. [18] applied histogram analysis to optimize brightness and contrast before using a random forest classifier to classify steel materials optical micrographs. Decost et al. [19] found it necessary to implement local HE before classifying additive manufacturing powder. Kitahara et al. [12] carried out adaptive histogram equalization (AHE) to classify steel surface defect micrographs. Ortegon et al. [20] performed normalization, HE and image denoising before classifying material phases based on SEM images. However, none of these studies provide specific details regarding the impact of preprocessing techniques on the performance of machine learning models. In addition, in other research fields such as natural image

and medicine, the color of images [21], size change methods [15], and normalization methods [22] have also been proven to affect the performance of machine learning. For material micrographs with significant differences in sample quality, reasonable preprocessing is more critical [23]. Regrettably, the impact of different image preprocessing techniques on machine learning based on such datasets remains unexplored.

In this work, we investigated the impact of different image preprocessing methods on the performance of machine learning models on micrograph datasets with significant sample differences for the first time. Especially, different types of tasks were taken into consideration. And the findings are expected to advance the understanding of image preprocessing steps within the materials community.

2. Methods

To effectively evaluate the impact of different image preprocessing methods on machine learning model performance, we utilized a four-step process: data collection, image preprocessing - I, image preprocessing - II, and neural network model, as illustrated in Fig. 1.

2.1. Data collection

To avoid excessive structural differences, we collected micrographs of Mg-Li based alloys from the published literature taken at similar magnifications and categorized them based on whether they exhibited deformation texture or not (the source references for micrographs can be found in the [supplementary file](#)). A total of 348 micrographs were included in the image dataset, with equal proportions of images in each category to ensure data balance. Specifically, the non-texture category includes as-cast optical micrographs of Mg Li-based alloys, while the texture category comprises optical micrographs of Mg Li-based alloys after rolling, extrusion, forging, and other deformation processes. The differences between images in the dataset as shown in Fig. 2, where these images differ significantly in structure, color, brightness, contrast and pixel intensity distribution. Additionally, we collected 170 data of tensile strength corresponding to the non-texture micrographs to complete the tensile strength prediction task. The distribution of the tensile strength after removing one outlier data (273Mpa) using the 3σ principle and box plot method is shown in Fig. 3.

2.2. Image preprocessing

We divided image preprocessing into two steps: step-I included image color and size adjustment, as well as normalization, while step-II focuses on enhancing the image quality.

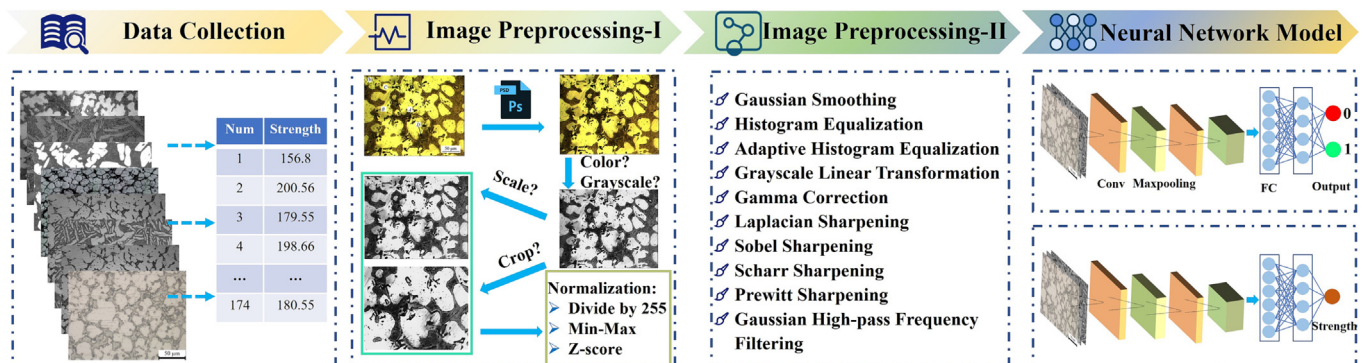


Fig. 1. Workflow for evaluating the impact of different preprocessing methods on the performance of machine learning.

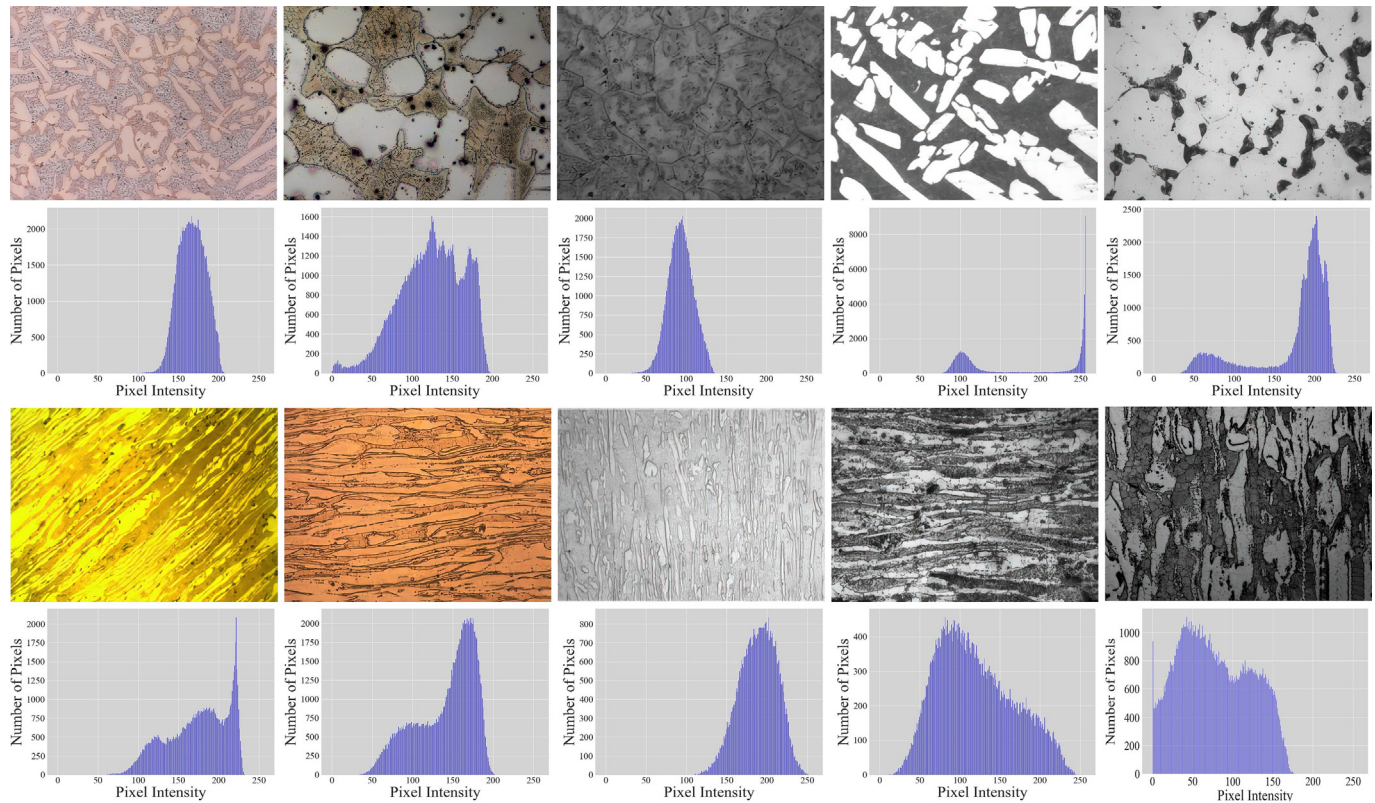


Fig. 2. Differences between images in micrograph datasets.

1) Color change

As the micrographs were taken from the published literature, they were often tagged with text or numbers, so Photoshop software was used to remove these confounding factors. Moreover, the micrographs were captured with different light sources, leading to variations in color. Petrushan et al. [24] demonstrated that color (RGB) and grayscale images produce different features. To examine the effect of color images versus grayscale images on machine learning, the original color images were converted to

grayscale. A comparison of the color and grayscale image is shown in Fig. 4. While the grayscale image lost color information compared to the color images, this did not appear to impact human expert analysis. However, it may have an unpredictable effect on machine learning models.

2) Size variation

Micrographs in the literature are often of varying sizes, which does not meet the requirement that the input to the model must be images of equal size. To address this, interpolation-based methods are commonly used to scale down or up images to meet model requirements [15]. However, Talebi and Milanfar [25] pointed out that the method used to change the size of images could affect the performance of the model. Therefore, we employed two frequently used and easy-to-implement methods to resize the images: interpolation scaling and center cropping. The results of two methods are presented in Fig. 5. When the original image size was smaller (Fig. 5-a) than the model input requirement (224×224), center cropping would fill the surrounding area of the image with 0 to meet the requirements (Fig. 5-b). When the original image size was larger than the model input requirement, the center crop would crop the image from the center of the image to meet the target size (Fig. 5-e).

3) Normalization

Image normalization is used to improve the convergence speed, accuracy, and stability of the model by changing the range and distribution of pixel values in the image [10]. The use of different methods of image normalization has been shown to affect the performance of machine learning [13]. The three commonly used methods of image normalization, including empirical method (i.e., dividing by the pixel intensity range value of 255), min-max

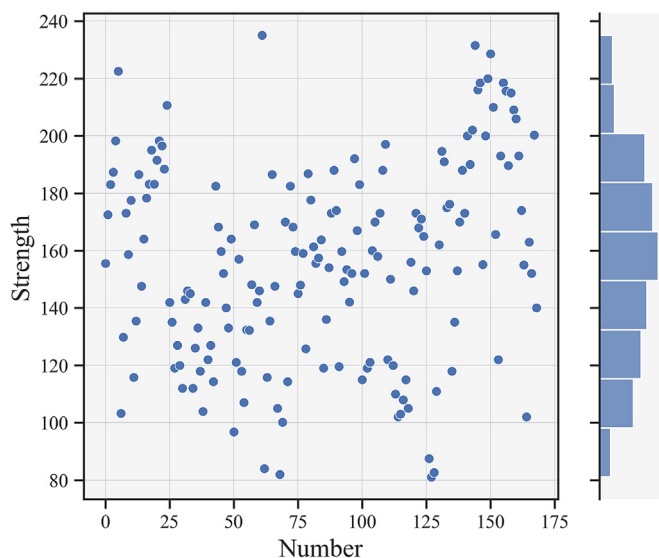


Fig. 3. Tensile strength distribution diagram.

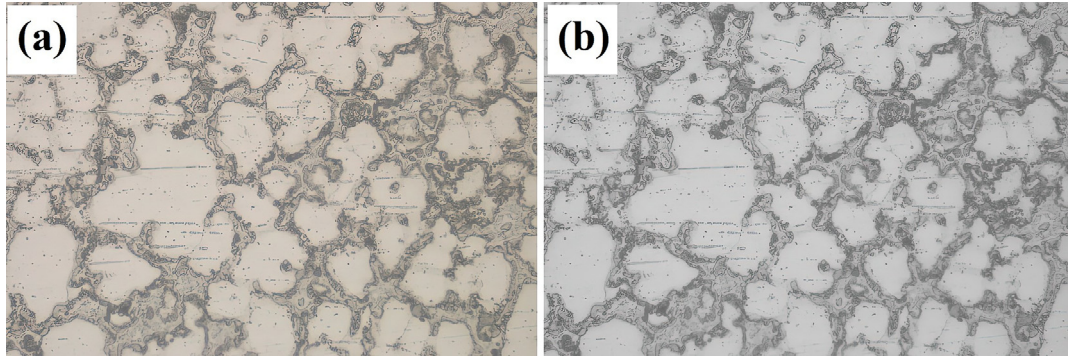


Fig. 4. A comparison of the color and grayscale image: (a) color image; (b) grayscale.

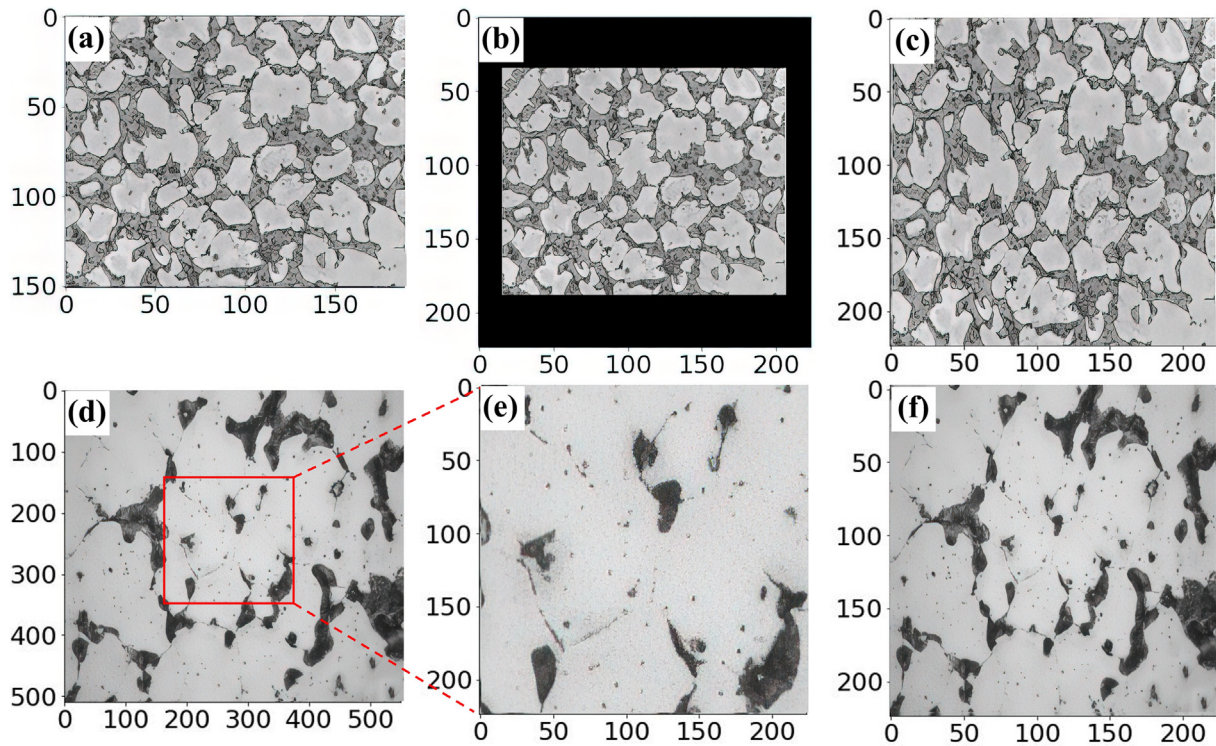


Fig. 5. Results of changing images of different sizes using central cropping and interpolation-based scaling: (a) and (d): original micrograph that is smaller/ larger than the required input size of the model; (b) and (e): center cropping; (c) and (f): interpolation scaling.

normalization, and Z-score normalization, were selected and compared. The min-max normalization method processes each pixel intensity according to Eq. (1):

$$x_{norm} = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (1)$$

In the above formula, x_i represents the pixel intensity value of the image, $\min(x)$ and $\max(x)$ represent the minimum and maximum values of the pixel intensity of the image, respectively, and x_{norm} represents the normalized pixel intensity. When the minimum value of the image intensity is 0 and the maximum value is 255, the min-max normalization is simplified as Eq. (2):

$$x_{norm} = \frac{x_i}{255} \quad (2)$$

Eq. (2) is an empirical method. Z-score normalization shifts and scales each pixel value in the image based on the mean and standard deviation of the dataset. The specific method of processing is shown in Eq. (3):

$$x_{norm} = \frac{x_i - \mu}{\sigma} \quad (3)$$

where μ is the mean of the image dataset (training set), and σ is the standard deviation of the image dataset. Fig. 6 shows the grayscale image and pixel intensity distribution after normalization using different methods. The original pixel intensity value of the image was between 0 and 255. Empirical method only scaled the pixel intensity of the image to the range of 0 to 1, without affecting the intensity distribution and visual effect. Min-max normalization also achieved intensity range scaling of the image, but made the intensity distribution of the image appropriately smoother, with a larger intensity range. Z-score normalization resulted in a significant change in the visual effect of the image because the program using matplotlib cannot display intensity values outside the range of 0–1 correctly, but the information stored in the image remained unchanged. Meanwhile, the intensity values of the image were also shifted to the range of –3 to 3.

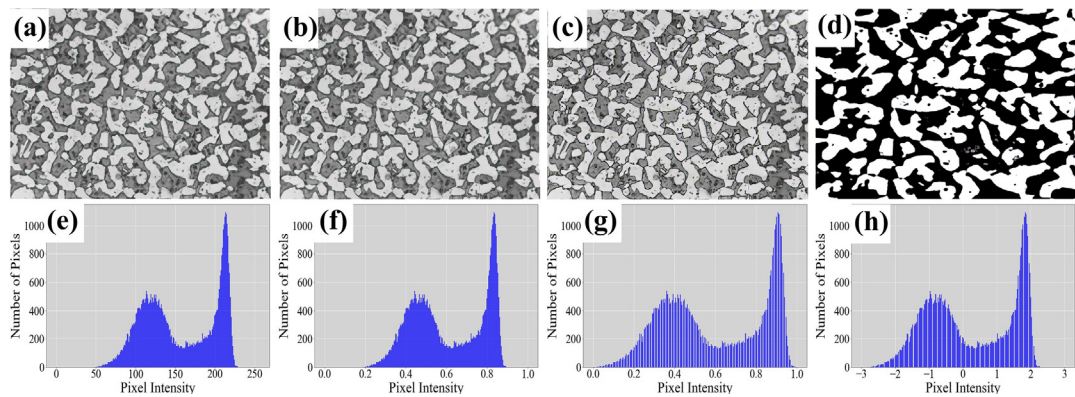


Fig. 6. The grayscale images and pixel intensity distributions after normalization using different methods: (a) and (e) original grayscale image and histogram before normalization; (b) and (f): grayscale image and histogram after empirical method; (c) and (g): grayscale image and histogram after normalization using min-max normalization; (d) and (h): grayscale image and histogram after normalization using Z-score normalization.

4) Image enhancement

The impact of noise [26], contrast [27], and sharpening [28] on image features has been well established. To enhance image quality, we performed color change, size change, and normalization, followed by Gaussian smoothing (GS), grayscale linear transformation (GLT), gamma correction (GC), HE, contrast limited adaptive histogram equalization (CLAHE), Laplacian sharpening, Sobel sharpening, Prewitt sharpening, Scharr sharpening, and gaussian high-pass filtering (GHPF). Table 1 presents the effects and parameters of each method used in our study. A detailed description of the image enhancement methods is provided in the [supplementary file](#).

The effects of different image enhancement methods (based on Fig. 6-a) are shown in Fig. 7, along with their respective histogram distributions. GS had no significant impact on the visual effect as well as on the histogram (Fig. 7-A and Fig. 7-a), suggesting the absence of obvious noise. However, this result was dependent on specific parameter settings. GLT led to contrast enhancement (Fig. 7-b) and reduction (Fig. 7 -c). Fig. 7(B) and Fig. 7(C) demonstrated that GLT can increase or decrease pixel intensity, resulting in a significant change in image brightness. Fig. 7(D) and Fig. 7(d) demonstrated the impact of GC with a gamma value of 0.5 on images and histograms. The lower grayscale regions (darker areas) of the image had stretched intensity values, which resulted in an overall brighter image, while the higher grayscale regions of the image were compressed into a smaller range. Conversely, when

gamma was greater than 1, the opposite effect occurred. HE, as shown in Fig. 7(F) and Fig. 7(f), expanded the gray levels that occupied a large proportion of the image, compressing those that occupied a smaller proportion, resulting in a more uniform distribution of pixels across the grayscale range. This enhanced the overall contrast of the image by increasing the dynamic range of gray value differences. In contrast, CLAHE (Fig. 7-G and Fig. 7-g) had a more uniform histogram distribution and produced a better visual effect than HE.

The techniques of Laplacian, Sobel, Prewitt and Scharr were all used for image sharpening through the detection of image edges. This study employed Laplacian, Sobel, Prewitt and Scharr operators for edge detection in the image and overlaid the detected edges on the original image (using a 0.5 overlay ratio) to achieve image sharpening. In the dataset used for this study, Sobel and Prewitt sharpening techniques produced more prominent effects with similar results, as previously demonstrated in Singh et al.'s research [29]. However, Laplacian sharpening was prone to producing unexpected artifacts in the presence of complex, heavily featured edges [30]. While Scharr sharpening could detect more edges [31], it may introduce too many detailed features. These two methods may require more precise parameter control to demonstrate better sharpening results. Fig. 7(I) illustrates the image after GHPF. GHPF enhanced high-frequency information in the image such as edge and detail information, but also reduced the overall brightness of the image.

Table 1
Role and main parameters of different image enhancement methods.

Method	Role	Main Parameters
GS	Remove noise and smooth image	Ksize = (3,3), sigmaX = 0 (based on OpenCV)
GLT -1	Enhance contrast and brightness	$Y = ax + b$, where x is the pixel intensity of the image, $a = 1.2, b = 0$
GLT -2	Reduce contrast and brightness	$a = 0.75, b = 0$
GC-1	Boost contrast and brightness and enhance dark details in images	$\gamma = 0.5$
GC-2	Reduce contrast and brightness, enhance detail in bright areas	$\gamma = 2$
HE	Adjust image brightness and improve contrast	Default parameters (based on OpenCV)
CLAHE	Improve local contrast	Clip limit = 2.0, tile grid size = (4,4) (based on OpenCV)
Laplacian sharpening	Highlight edge features in images and enhance detail	Ksize = 3 (based on OpenCV)
Sobel sharpening/Prewitt sharpening/Scharr sharpening	Highlight edge features in images and enhance detail	Default parameters (based on OpenCV)
GHPF	Highlight edge features in images and enhance detail	Cut-off frequency = 30

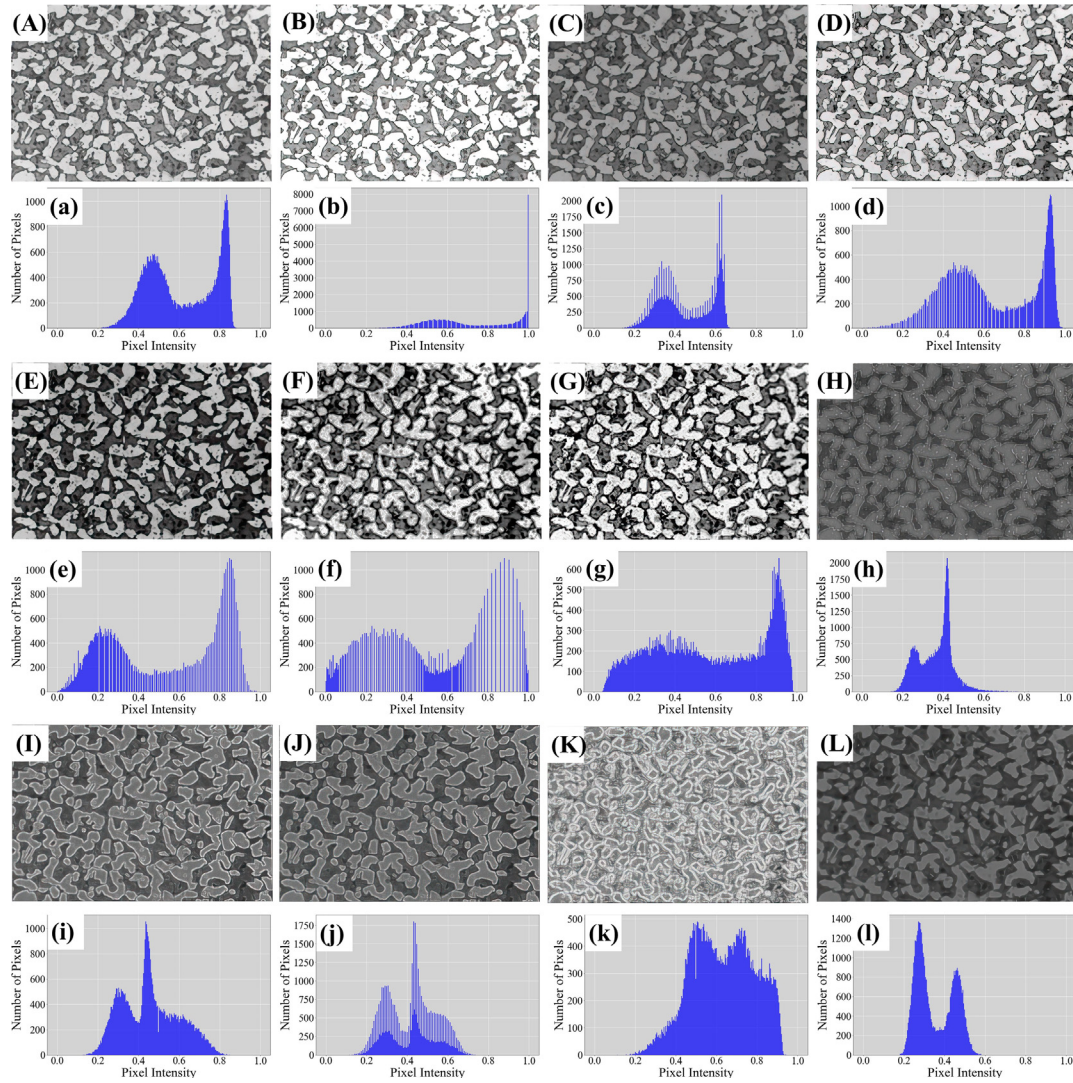


Fig. 7. Images and histograms after different image enhancement algorithms: (A) and (a): GS; (B) and (b): GLT-1 ($a = 1.2$, $b = 0$); (C) and (c): GLT-2 ($a = 0.75$, $b = 0$); (D) and (d): GC-1 ($\gamma = 0.5$); (E) and (e): GC-2 ($\gamma = 2$); (F) and (f): HE; (G) and (g): CLAHE; (H) and (h): Laplace sharpening; (I) and (i): Sobel sharpening; (J) and (j): Prewitt sharpening; (K) and (k): Scharr sharpening; (L) and (l): GHPF.

2.3. Neural network model

Compared to traditional machine learning methods like random forests and support vector machines, convolutional neural network (CNN) has broader applications in image recognition and classification [32,33]. CNN can directly extract image features without relying on complex domain knowledge. In this paper, we employed the transfer learning approach by using the feature extraction layer of the pre-trained VGG16 model to extract features from images pre-processed in different ways. Then, we reconfigured the prediction layer (i.e., fully connected layer) of VGG16 to accomplish image classification (texture or not) and tensile strength prediction tasks. VGG16 has demonstrated excellent accuracy on the ImageNet dataset [34] and possesses the capability to extract critical features from images. The modified VGG16 model used in this paper is illustrated in Fig. 8. The implementation process of modified VGG16 model is provided in the [supplementary file](#). Before model training, the image dataset was divided into training set and testing set. To enhance the generalization performance of model, data augmentation techniques, such as clockwise rotation by 90 degrees and diagonal flipping, were applied to the classification dataset. Other parameters for the modified VGG16 classification and

regression models are displayed in Table 2 and Table 3, respectively. The classification model employed the cross-entropy loss function as the loss function, with the calculation formula shown in Eq. (4):

$$L = -\frac{1}{n} \sum_i \sum_{c=1}^M y_{ic} \log(p_{ic}) \quad (4)$$

where M represents the number of categories, y_{ic} is the indicator function (0 or 1), taking 1 if the true category of the i -th sample is c and 0 otherwise, p_{ic} is the predicted probability that the i -th sample belongs to category c , and n is the number of samples. The regression model used the mean square error (MSE) as the loss function, and the detailed calculation formula is shown in Eq. (5):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 \quad (5)$$

where y_i and $f(x_i)$ respectively represent the true value and predicted value of the i -th sample. The coefficient of determination R^2 was used as an indicator to evaluate the generalization ability of the strength prediction model. The calculation formula of R^2 is given by Eq. (6):

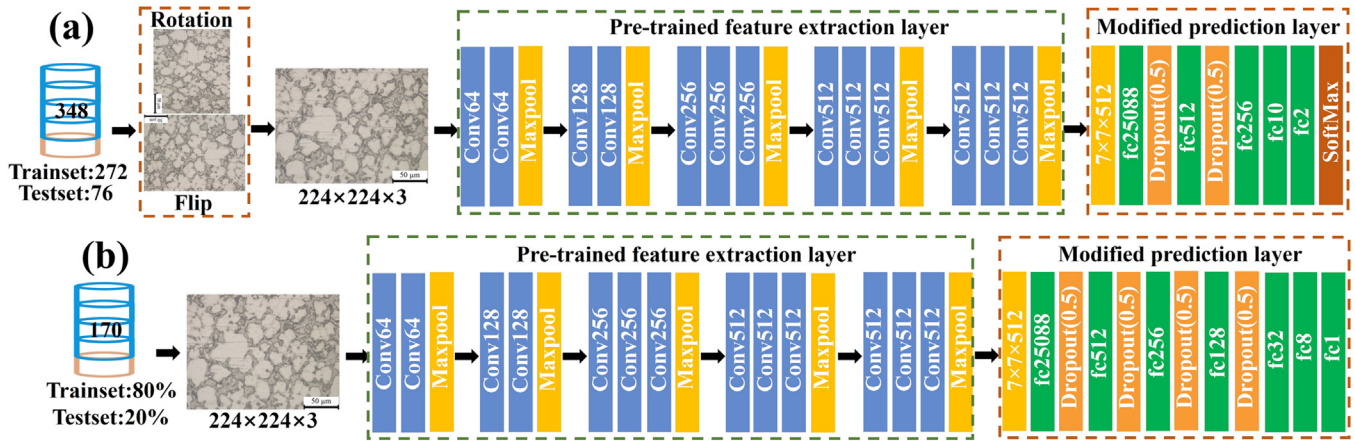


Fig. 8. Modified VGG16 model: (a) Model for image classification; (b) Model for predicting tensile strength.

Table 2

Other parameters of the modified VGG16 model for image classification.

Parameter name	Value
Optimizer	Adam
Learning rate	0.0001
Epoch	100
Train batch size	32
Early Stop Patience	7
Loss function	Cross Entropy Loss

Table 3

Other parameters of the modified VGG16 model for predicting tensile strength.

Parameter name	Value
Optimizer	Adam
Learning rate	0.0001
Epoch	350
Train batch size	16
Early Stop Patience	7
Loss function	Mean Square Error

$$R^2 = 1 - \frac{\sum_{i=1}^n (f(x_i) - y_i)^2}{\sum_{i=1}^n (\frac{1}{n} \sum_{i=1}^n y_i - y_i)^2} \quad (6)$$

3. Results

3.1. Task 1: Image classification

3.1.1. Results of classification model after different preprocessing-I

Fig. 9 illustrates cross-entropy loss and the accuracy on testing set of our classification model when different resizing methods were applied to color and grayscale images as inputs. After selecting the image color and modifying the image size, we only used the original image intensity values and divided the image intensity by 255 as inputs to our model. These two methods did not compromise the image quality. Our findings indicated that the additional color information in RGB color images did not significantly improve the classification accuracy compared to grayscale images. The center crop method outperformed direct scaling of the image, irrespective of the image color or the range of pixel intensity. The model achieved the highest classification accuracy (90.79%) by performing center crop and pixel intensity divided by 255 on gray-

scale images. Consequently, we adopted grayscale images and center crop as preprocessing steps before normalization.

Fig. 10 displays the cross-entropy loss and accuracy on testing set of the model using different normalization methods. Dividing pixel intensity by 255 yielded the highest accuracy, while Z-score

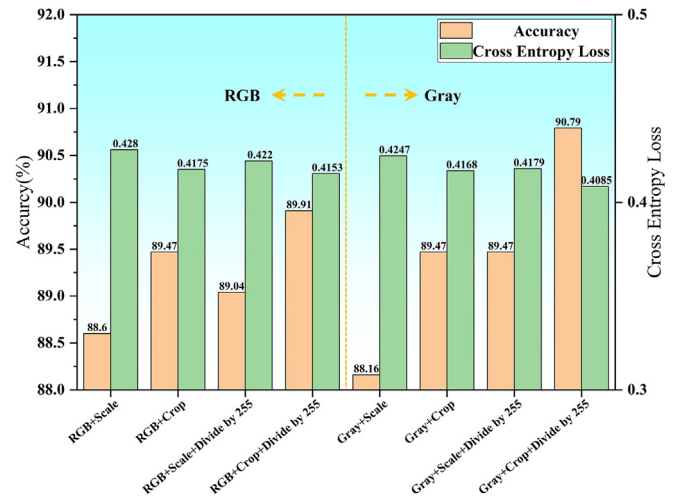


Fig. 9. Accuracy and cross-entropy loss of classification models under different image color and size change methods.

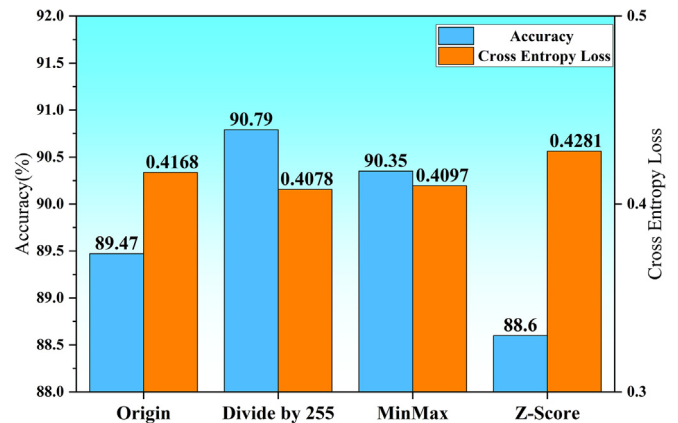


Fig. 10. Accuracy and cross-entropy loss of the classification model after applying different normalization methods.

normalization yielded the lowest. This could be attributed to Z-score normalization calculating the mean and standard deviation of the entire dataset to normalize each image, potentially leading to some images with large deviations from the mean losing significant detail. On the other hand, normalizing pixel intensity by dividing by 255 and min-max normalization processed each image based on its pixel intensity characteristics, retaining as many features as possible. From Fig. 10, normalizing pixel intensity by dividing by 255 appeared more suitable for micrographs with significant image differences.

3.1.2. Results of classification model after different preprocessing-II

Fig. 11 presents the cross-entropy loss and accuracy of the classification model using different image enhancement methods. Not all image enhancement techniques had a positive effect on the model. Among the methods tested, HE had the greatest impact on improving the model's accuracy, reaching a classification accuracy of 93.42%. Existing literature has demonstrated the applicability and excellent performance of HE in enhancing micrographs of materials [10,19]. In contrast, GLT and GC resulted in only a slight improvement in classification performance. As expected, Laplace

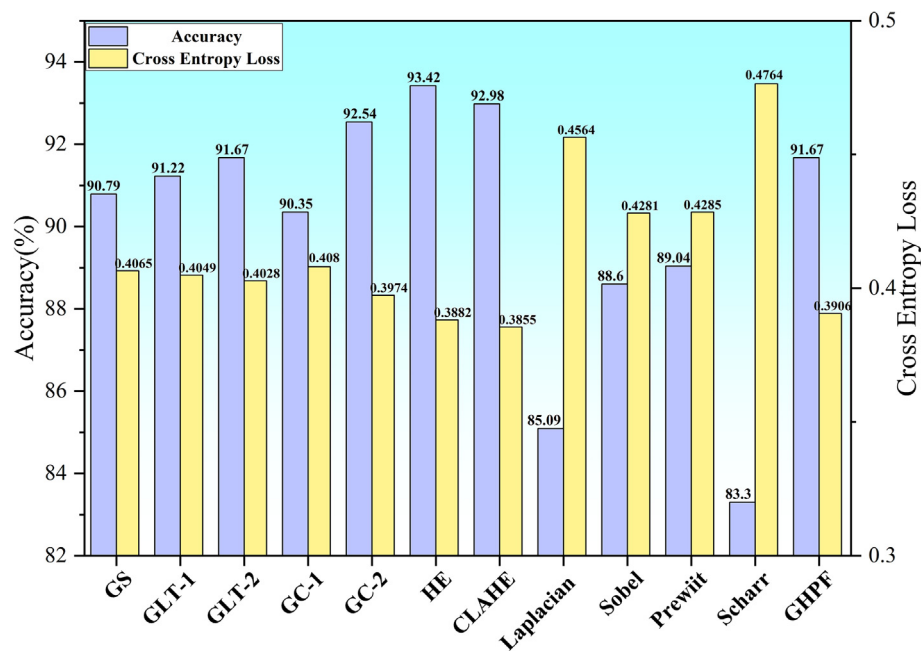


Fig. 11. Classification accuracy and cross-entropy loss of the model after applying different image enhancement methods.

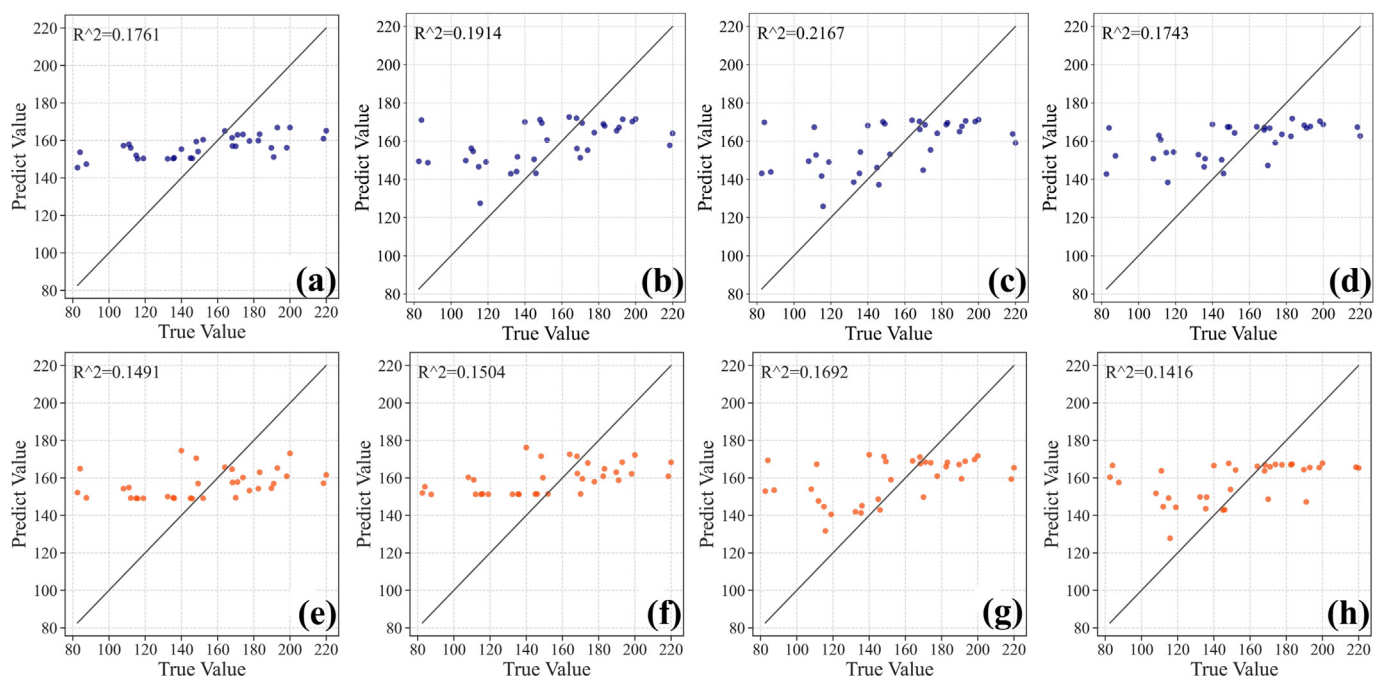


Fig. 12. Prediction results of tensile strength after different image size change methods and normalization methods: after center cropping, (a) origin; (b) divided by 255; (c) min-max; (d) Z-score; After directly scaling the size, (e) origin; (f) divided by 255; (g) min-max; (h) Z-score.

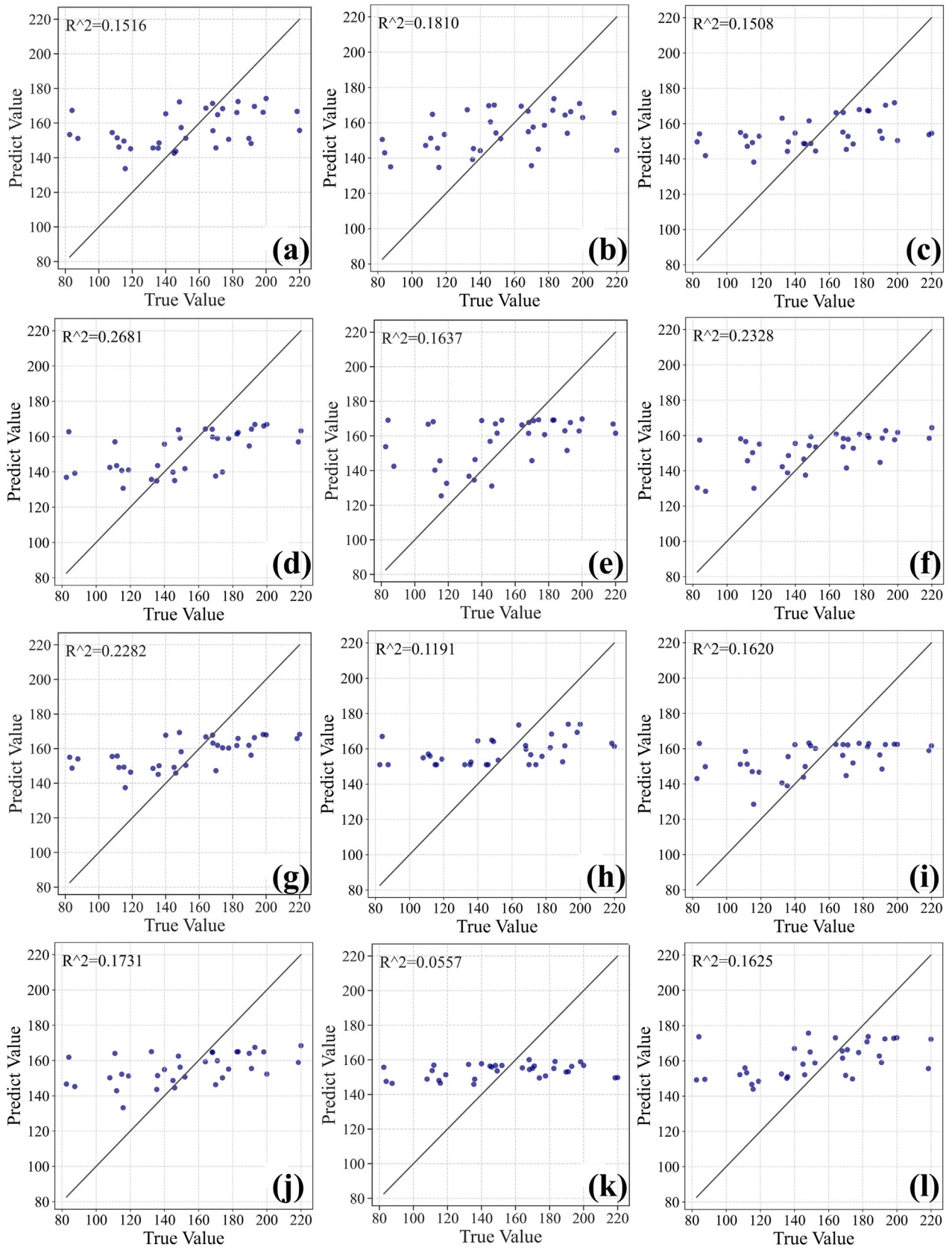


Fig. 13. Prediction results of tensile strength using different image enhancement methods: (a) GS; (b) GLT -1; (c) GLT -2; (d) GC -1; (e) GC -2; (f) HE; (g) CLAHE; (h) Laplace sharpening; (i) Sobel sharpening; (j) Prewitt sharpening; (k) Scharr sharpening; (l) GHFP.

sharpening and Scharr sharpening had a negative impact on the model, possibly due to introducing additional noise and excessive enhancement of edge features leading to decreased accuracy. Similarly, Sobel sharpening and Prewitt sharpening failed to produce expected improvements, indicating that prominent edge features were not suitable for the micrograph dataset. Additionally, GHFP produced a minor improvement in performance. For the classification task in this study, the high-frequency information in images could potentially have a beneficial impact on the classification model. However, excessive enhancement should be avoided.

3.2. Task 2: Tensile strength prediction

3.2.1. Results of regression model after different preprocessing-I

In Section 3.1, we found that color images had similar classification performance to grayscale images on our dataset. As a result, we processed only grayscale images in the regression task. Fig. 12 displays the results of our tensile strength prediction model using various image size change and normalization methods. It was observed that the model trained on the center cropped dataset had a higher R^2 value compared to directly scaling the image size. Additionally, using normalization methods was found to be necessary, as opposed to directly using the original intensity values of the image. Among the different normalization methods, min-max normalization achieved the best performance for the model after applying various image size change methods. This differed from the best normalization method used in classification task, suggesting that different tasks on the same dataset had unique requirements for normalization methods. Therefore, center cropping and min-max normalization were more suitable for the dataset and tensile strength prediction task in this article. As a result, subsequent image enhancement would be performed based on these approaches.

Fig. 13 displays the results of the predicted tensile strength after using different image enhancement methods. Unfortunately, only GC, HE, and CLAHE showed positive results. Among these methods, the model with GC achieved the highest determination coefficient R^2 of 0.2681, while HE and CLAHE had little improvement on the model. Image sharpening was found to be unsuitable for enhancing images in strength prediction model, as the model with Scharr sharpening had the lowest R^2 of 0.0557. While our regression model could not accurately predict strength as micrographs features alone cannot fully capture the predicted strength features [6,10,35,36], our results suggested that the features extracted from micrographs after different preprocessing had varying degrees of correlation with material strength.

4. Discussion

4.1. Effect of image color on model performance

Color does not appear to be a significant feature for accurately classifying microscopic images. Bui et al. [37] demonstrated that models trained on grayscale images have higher classification accuracy than those trained on RGB images. Conversely, Singh et al. [38] emphasized the importance of color in classification tasks for different datasets such as Oxford-Flowers and Oxford-IIIT Pets. Xie et al. [21] suggested that the selection between grayscale and color images should be based on the dataset characteristics.

In metallic materials, different metal grains or phase structures affect light absorption, scattering, and reflection, resulting in varying pixel intensity values. These intensity values typically appear in grayscale rather than color. Differences in intensity values allow for visual differentiation between phases or structures, including

shape and texture information. These intensity values are primarily influenced by the chemical composition and lattice structure of the phases and can be recognized by machine learning models. The various hues in the microstructure images present in this paper are a result of differences in the light source and filters used by the imaging equipment, rather than conveying true color information. Nevertheless, this conclusion can also be extended to true color microstructure images.

4.2. Impact of size change methods on model performance

Center cropping showed better performance on various tasks in our dataset. Direct scaling can alter the image shape to some extent as shown in Fig. 5(c) and Fig. 5(f), with the degree of change depending on the original and target sizes of the image. The classification task in this paper is to determine whether the structure in the image is texture. Texture is a phenomenon in which the grains of polycrystalline materials tend to align in a particular orientation under external conditions [39,40], and shape is an important factor in determining texture. The shape of different phases or structures can also impact the material strength [41,42], which may explain why center cropping could also perform well in predicting material strength. While center cropping cannot preserve the global information of microstructure images, these images are typically carefully selected by authors in the literature to represent the overall structural information of the material. Uniform distribution of structure is a key feature of such images [43,44]. Meanwhile, Fig. 14 shows the size distribution of the micrograph dataset. The size distribution of the image is concentrated between 200 and 400, indicating that the information loss caused by using center cropping is acceptable. Therefore, the cropped image can represent global information to some extent, its effectiveness depends on the size difference before and after cropping. Additionally, some images in the dataset that are smaller than the required size are zero-padded as shown in Fig. 5(b). Hashemi [15] demonstrated that zero-padding does not impair the accuracy of the model and could even reduce its training time.

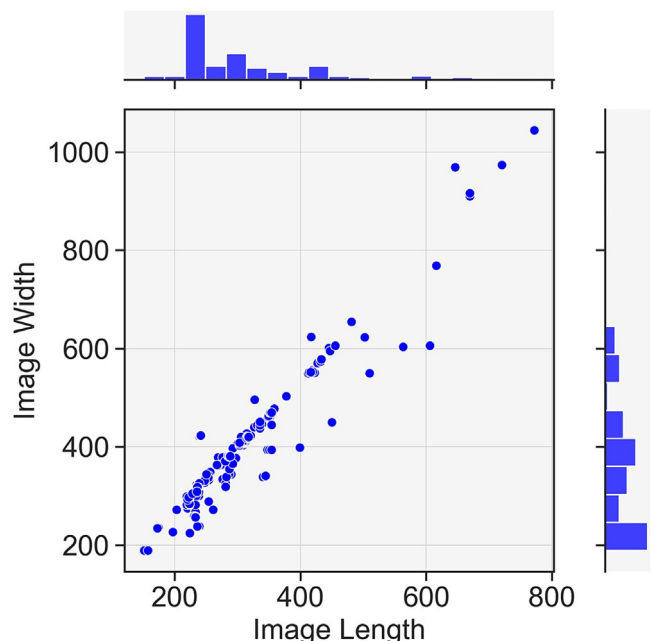


Fig. 14. The size distribution of the micrograph dataset.

4.3. Impact of normalization on model performance

Raad et al. [45] suggested that the effectiveness of different normalization techniques in improving model performance depends on the dataset characteristics. Similarly, Singh et al. [22] have shown that there was no universally best normalization method across different datasets. In this study, we find that normalizing pixel intensity by dividing it by 255 resulted in higher accuracy in classification tasks, whereas min-max scaling performs better in predicting tensile strength. The efficacy of normalization techniques is specific to the task at hand. Notably, Z-score normalization proves inadequate for the micrograph dataset used in this paper. Significant differences among the images in our dataset mean that normalizing individual image using Z-score can alter their original characteristics and potentially decrease model accuracy. Hence, image normalization is task and dataset-dependent, and specific guidelines for using particular methods cannot be provided.

4.4. Impact of image enhancement methods on model performance

In classification task, HE provides the most significant improvement in model accuracy. HE effectively enhances image contrast [46], which is closely linked to higher accuracy in our results. Moreover, GLT and GC can adjust image contrast and also enhance the model performance to varying degrees. However, they are more sensitive to the influence of the original image intensity distribution. Different degrees of enhancement by using GLT and GC are produced due to the significant differences in intensity distribution of each image in the micrograph dataset used in this study.

CLAHE is often regarded as a superior image enhancement technique compared to Histogram Equalization. However, it does not yield optimal results. One potential explanation for this discrepancy is that CLAHE may excessively amplify some noise and classification-related detail information [47], which could hurt the model performance. Fig. 11 illustrates that the cross-entropy loss of CLAHE is lower than that of HE. This suggests that CLAHE introduces more features that impact the model classification, which may lead to high confidence in correctly classifying specific images but can also lead to the classification wrongly of other images. This limitation can potentially be alleviated through further parameter adjustments. Similarly, parameter adjustments can further enhance the effects of GLT and GC. Nonetheless, we believe that HE remains an excellent default choice for enhancing microscopic images, given its simplicity and effectiveness.

In regression task, the model trained on the dataset processed with GC-1 achieves the highest coefficient of determination. Obviously, image enhancement also displays task-specific effects. When $\gamma < 1$, gamma correction can enhance the low-intensity regions of an image, resulting in an overall increase in brightness (Fig. 7-D). While GLT can also alter image brightness, the change is more radical (Fig. 7-B and b). The GC-1 processed images have better visual comfort and intensity distribution. In contrast, HE and CLAHE produce less significant brightness enhancement, especially in the lower gray level areas [48]. GC can better adjust the brightness while adjusting the contrast [49]. Zhong et al. [10] demonstrated that brighter micrographs are advantageous for prediction ultimate compressive strength of molecular solid materials, although this may not be the case for all models. Nevertheless, our results show that on our dataset, brighter images have a primary affinity for tensile strength prediction, followed by contrast. The performance of GHPF in different tasks shows that image classification is sensitive to contrast, while tensile strength prediction has a somewhat positive correlation with brightness.

Image sharpening has negative effects on both tasks, despite the good visual distinction effects of Sobel and Prewitt sharpening. For

our dataset, image sharpening may be overly enhanced, leading to the masking of other important features [50].

We must also acknowledge some limitations of our work. Our study only focuses on one frequently used machine learning model, and different models are sensitive to image properties [10,51]. This will be our next focus of consideration. Moreover, the combination of multiple processing methods may result in unexpected outcomes. As there are numerous combinations, a more sophisticated design may be necessary, rather than relying on exhaustive methods.

5. Conclusion

In this study, we explored for the first time how different image preprocessing techniques impact machine learning performance for classification and regression tasks on a micrograph dataset with significant sample differences. Color features were not necessary for the model. The neural network model was sensitive to the method of resizing images, and center cropping performed better in different tasks while without altering the shape information. The normalization method used for images should be selected based on the task and dataset characteristics. Dividing image intensity by 255 had a better improvement effect on the model for classification tasks, while min-max normalization was more suitable for predicting tensile strength. Histogram equalization had the most significant improvement effect on image classification models, while gamma correction was more appropriate for in tensile strength prediction. The classification model was more responsive to adjusting the contrast of images, whereas the tensile strength prediction model tended to benefit from a proper increase in image brightness. Image sharpening were not suitable for micrograph datasets with large sample differences. Overall, this research contributes to aiding researchers in selecting appropriate preprocessing methods for their specific tasks and datasets.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

A subset of the micrograph dataset for this study can be found at: <https://aistudio.baidu.com/aistudio/datasetdetail/222788>.

Code availability

Code used for these findings is available upon reasonable request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matdes.2023.112086>.

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